

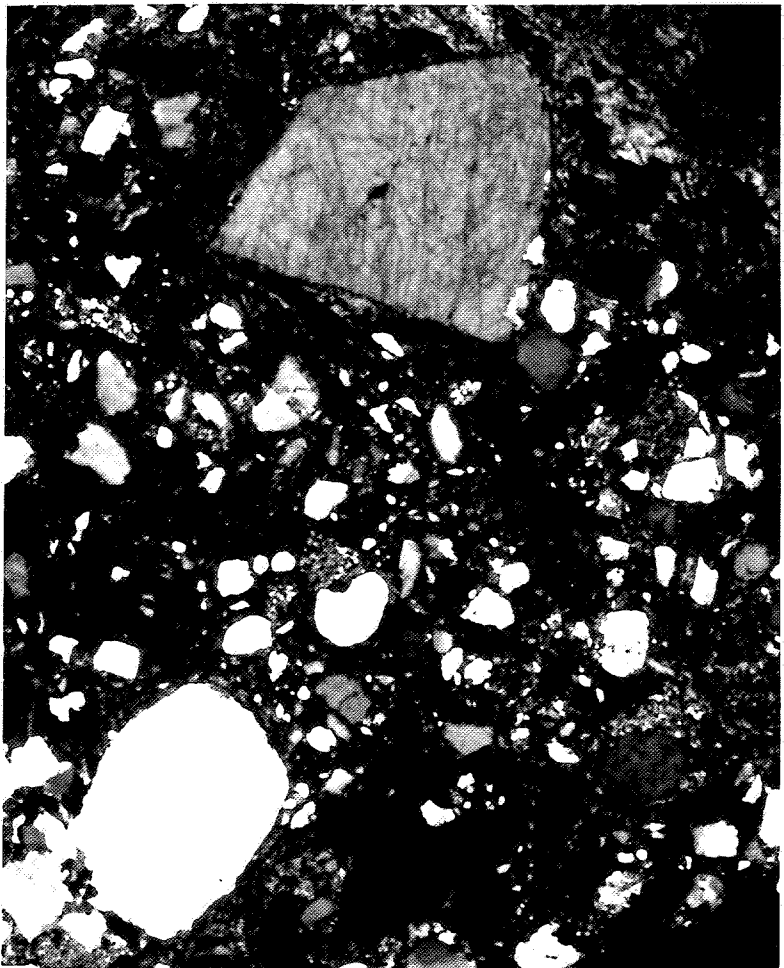


Figure 2

Sandy clay from Elizabeth Creek thought to be Tregolana shale but more likely a much later deposit. (70742/A9861).

Many rounded quartz grains and few small ferruginous cherts (speckled black and white) are probably derived from Whyalla Sandstone. More angular grains may have derived from pandurra Quartzite. Rock matrix is mainly of clay mineral (grey and white stipple); secondary chert (dark areas), replacing clay. Scattered, fine quartz grains (white) are also seen.

Crossed nicols

Magnification X 15

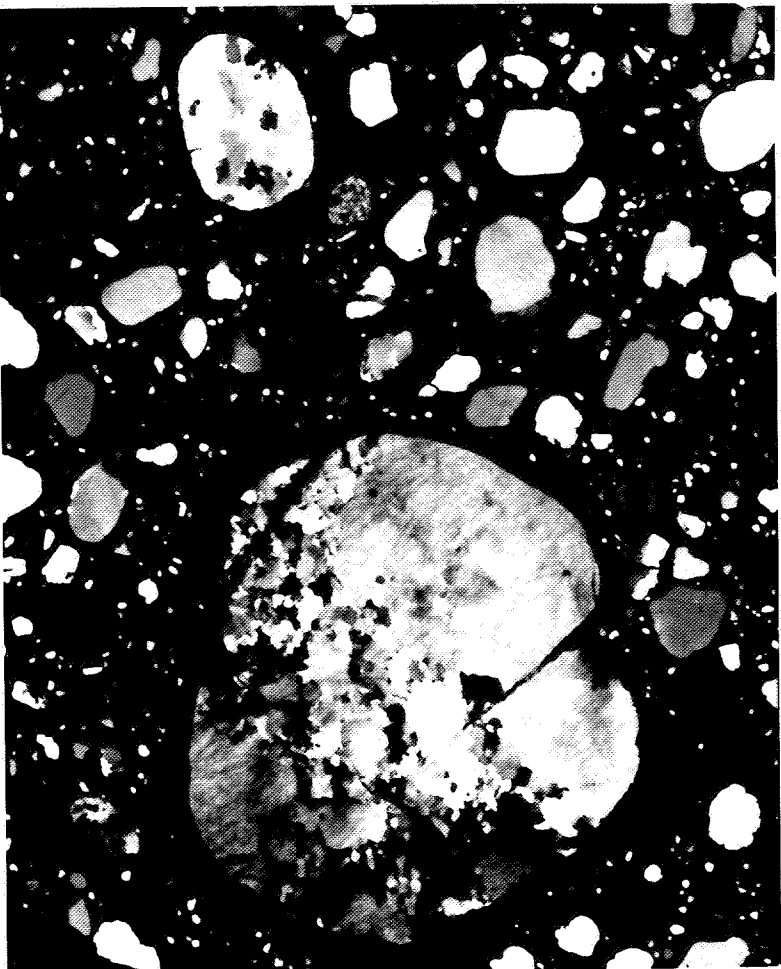


Figure 3
Ferruginous deposit from Elizabeth Creek (70743/
A9862).

Large round quartz rock grains (grey and white mosaic patterns) and quartz mineral grains (flat white and grey) are seen. Ferruginous cherts (speckled black, white and grey) are present. Several rounded cherts are obscured by iron staining. The groundmass is largely of fine limonitic quartz (black) and fine clear silt size quartz grains (white). Whyalla Sandstone.

Crossed nicols

Magnification X 15

A PETROGRAPHIC/MINERAGRAPHIC EXAMINATION
OF SOME MT. GUNSON ORE SPECIMENS
FROM THE WEST LAGOON AND MAIN OPEN PITS

PACMINEX PTY LTD

Dated in Sydney
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TABLE 1 MT. GUNSON COPPER ORES - CHEMICAL ANALYSES

TABLE 2 MT. GUNSON COPPER ORES - MINERAL COMPOSITIONS
(CALCULATED FROM CHEMICAL ANALYSES)

1. INTRODUCTION

During a visit to Mount Gunson between the 15th and 16th July of this year a number of specimens were taken of Red Quartzite and Lithic Sandstone near the top of the Pernatty Grit. Quarries visited were the West Lagoon which is presently being mined for copper sulphide ore and the Main Open Cut an abandoned mine in oxide copper ore. Collection points for the ores were marked on plans. Some specimens from the Main Open Cut (Nos. 7 and 8) were mislaid and these will have to be resampled if possible.

1.1 Sampling

Co-ordinates are given indicating sampling points, 2 to 6 from the West Lagoon pit and 9 and 10 from the Main Open Cut.

Mineralisation at the West Lagoon Pit was in the form of sulphides - chalcopyrite, chalcocite, covellite and some bornite and digenite which form the cement of friable sandstone up to a thickness of 14 feet and mainly at the junction between the two formations of Red Quartzite below and Lithic Sandstone above.

Similarly, mineralisation in the Main Open Cut occurs at the junction of the Red Quartzite and Lithic Sandstone formations and is mainly malachite or chrysocolla which are frequently intergrown.

2. DISCUSSION

The work done to date on the Gunson deposits can at best be regarded as only preliminary and at present consists of petrographic examination of selected specimen from the Pernatty Grit of the West Lagoon and Main Open Pit. Some samples from the latter mine were unfortunately lost and sampling from the former was a little bit hectic owing to the heavy rain which made the bottom of the pit something of a

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quagmire.

However some ideas on the petrogenesis of Gunson ores have come out of the petrography and are related below, dealing firstly with the West Lagoon Pit (sulphides) and then the Main Open Pit ('oxides').

- Whyalla*
- i) Most Lithic Sandstone samples consist of large very rounded quartz grains (av. 1 mm) with tiny silt-size quartz grains and clay as a more or less continuous matrix. Red Quartzite consists ^{*Pandana*} mainly of poorly sorted quartz grains, subangular to rounded and much lithified; with clay sporadically in pockets throughout.
- ii) The sediments are presently mainly lithic arenite while some specimens approach lithic wacke (as per Williams, Turner and Gilbert) and have suffered much diagenetic alteration. This applies particularly to the Red Quartzite where many of the pores between primary quartz grains have been filled by secondary quartz. This has replaced clay cement but not relict limonite trains which outline and indicate the roundness of many of the original quartz grains. Diagenetic growth of the quartz grains now give them an appearance of unusual angularity but most grains were originally rounded. Addition of secondary quartz to original grains does not change their optical continuity.
- iii) Sulphides, silt and clay as seen in the Lithic Sandstone were dumped into flood channels with coarse sand as described in Section v). Later diagenesis led to recrystallisation and nucleation of sulphides into ribbon-like network forms in the interstices. Some growth of quartz sand grains and limited replacement of sand grain margins and clay mineral matrix can be seen.
- iv) The presence of framboidal pyrite in the clay matrix of sample 3A and of pyrite atolls around chalcopyrite grains in sample 2A indicate biochemical precipitation of primary sulphides in a sapropelic environment such as deposition in the bottom muds

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in a closeby playa lake or lagoon and is at odds with the irregular coarse to fine grit and sandstone deposition of the main rock fabric. Coarse sands, and poorly sorted sands at that, were either deposited in deep water by turbidity currents or in a shallow tectonically unstable shelf environment in torrential channels adjoining the littoral. The presence of cross-bedding rather than graded-bedding in the sediments would suggest the latter type of deposit. Calcium sulphate forms in playa lakes behind sea shore lines and normally under stagnant conditions. Metals in solution from eroding rocks in the catchment area would accumulate in such a stagnant environment and with the aid of sulphate reducing bacteria hydrogen sulphide was produced utilising the calcium sulphate and precipitating metal sulphides which quite rapidly accumulated on the bottom of the 'lake'. Since the bacteria required some organic matter for their metabolism this could well have been supplied by debris from marginal algal reefs which are known to have occurred in adjoining Woocalla Dolomite of similar age. Seasonal floods occur from time to time in desert environments such as this, partly washed out the playa lakes and carried clay/silts and sulphides mixed-up with coarse sands caught-up by the flood waters into the flood channels, the sand mixing with the kaolin matrix of the Red Quartzite and the kaolin and silt matrix of the Lithic Sandstone. The generally rounded shape of the primary grains, lack of heavy minerals other than zircon and tourmaline and lack of feldspar indicate an already formed sandstone as the source of sand grains in the channel deposits, not a primary igneous source. Subsequent burial of the deposits through subsidence encouraged mobility under supergene conditions of sulphides and deposition of the 'cement' sulphides just below the water table. Movement of sulphides and some silica appears to have taken place about the same time, since in at least one specimen supergene chalcocite is seen to be associated with secondary quartz in a 'vein'. Primary copper iron sulphides appear to have broken down under supergene conditions to give complex covellite/bornite/digenite intergrowths. The final sulphide supergene product is chalcocite which frequently

occurs in coarse veins and fissures particularly seen in the Red Quartzite. Covellite occurs disseminated in Lithic Sandstone and possibly pseudomorphs original chalcopryrite in an environment where there is restricted movement of solutions. Alternatively where chalcocite and covellite occur as disseminated grains these may have been primary precipitates in the playa lake environment, particularly where the sulphide is seen to be finely intergrown with clay mineral, the clay particles acted as a nucleus for precipitation of sulphides.

- v) In view of the proximity of sulphide deposits in rocks of equivalent age it is considered unlikely that oxide copper minerals would have been deposited concurrently with sulphides particularly since original environmental conditions in host rocks do not appear to have been noticeably different. Also the inhomogeneous nature of crystallisation and mineral replacement suggest supergene rather than primary precipitation of basic copper carbonate and silicate as considered possible by King (Ref. 6). No relict sulphides or sulphide textures were seen in specimens selected but specimens of relict sulphide have been found in the Main Open Pit. Deposition of primary sulphides in gel form along fractures and joint planes in formed Lithic Sandstone and Red Quartzite just above a rapidly receding water table would lead to rapid oxidation of the sulphides by oxygenated CO₂ charged water. An enriched supergene sulphide horizon may exist below the oxidised levels, but with the absence of leached zones in the oxide ores it seems unlikely.

- vi) The basic copper chlorides, atacamite and paratacamite were not strictly identified in thin section since the mineral, thought to be atacamite showed unusual optical properties. However X-ray diffraction examination by Dr. Deleon of the C.S.R. Research Laboratories confirmed the presence of atacamite generally replacing malachite in specimen 9 and from specimens taken in the metallurgical 'test area' of the Main Open Pit.

very important

- vii) The clay matrix of unmineralised Lithic Sandstone and Red Quartzite was examined by X-ray diffraction by C.S.R. Research Laboratories and was confirmed as kaolinite.
- viii) Total copper, acid soluble copper (chalcocite test), total iron and some aluminium determinations have been done on samples 1, 2A, 4, 5, 5A, 6A, 6B and 6C all from the West Lagoon Open Pit. Total copper, acid soluble copper (chrysocolla), acid soluble copper (malachite), chloride, carbonate and phosphate have been determined on samples 9B and 10 from the Main Open Pit.

The calculated mineral composition from assay frequently conflicts with the approximate modal composition as observed in section.

3. SUGGESTIONS FOR FURTHER WORK

- i) To collect bornite from the West Lagoon Pit to study its paragenetic relationships with other sulphides and the gangue present.
- ii) To collect more chalcopyrite from the West Lagoon Pit to study further its relationships with the framboidal pyrite present.
- iii) To collect specimens of known atacamite and paratacamite from the Gunson area.
- iv) To see maps showing sedimentary features such as load casts, ripple marks, washouts and direction of false bedding, etc.
- v) To do some limited sulphur isotope work on selected sulphides if this has not already been done.
- vi) To see tailing sample from the Mill after flotation to study sulphide losses.

4. RESULTS

4.1 Petrography

4.1.1 West Lagoon

Samples 2A Lithic Sandstone and 2B Red Quartzite taken from adjoining positions NW of the Open Pit.

Specimen 2A (20867mN 6766mE) - In view of clay matrix, most likely a Protoquartzite (as per Pettijohn). WHYALLA SANDSTONE

The section consists of rounded to subangular quartz grains which are almost equigranular (av. size 200μ). Diagenesis of the quartz component has added quartz (which is optically continuous with primary quartz) to grains, excluding the original clay mineral cement and producing a silica cemented patchy quartzite. However the excluded clay has been squeezed into pockets as large as some of the smaller quartz grains. After very careful checking of the section no identifiable feldspar was seen. Occasional large purple detrital zircon grains and greenish tourmaline prisms were seen in the matrix. Trains of fine limonite particles just inside quartz grains forming the fused mosaics indicate original position of quartz grain boundaries. With the quartz grains are seen occasional 'quartz gneiss' grains of tiny mosaics of sheared quartz grains.

Sulphides present are bands of chalcopyrite > chalcocite. The chalcopyrite forms segregations between quartz grains where crystal growth has not sealed off the quartz mosaics, i.e. where a clay matrix remained between the quartz grains. The chalcopyrite, generally 10 to 20μ av., maximum 100μ , fills completely spaces between the primary quartz grains and contains marginal discontinuous pyrite relicts (atoll textures indicating syngenetic, simultaneous crystallisation of the sulphides) with occasional tiny quartz inclusions giving the appearance of having replaced part of the clay matrix. The rounded appearance of the pyrite is reminiscent of framboidal pyrite seen in specimen number 3A.

Sphalerite to 0.3 mm is seen with continuous margins (15μ) of chalcopyrite boundaries suggest simultaneous crystallisation and rejection of chalcopyrite by sphalerite to grain boundaries = atoll texture. The sulphides have replaced the 'clay' from between quartz grains. The replacement of clay by chalcopyrite is diagenetic and post depositional.

Flecks of chalcocite (av. 20μ , maximum $1,000\mu$) occur sporadically in one corner of the section. These are not seen associated with other sulphides but as material which has replaced interstitial clay.

A rough modal analysis gave the following v/v percent results. Quartz 95, clay mineral 3, zircon + tourmaline + sphene 1, sulphide <1.

PANDURRA FM.

Specimen 2B (20867mN 6766mE) - Red Quartzite. Similar to Lithic Sandstone, stratigraphically just above, but coarser grained, the grains being more irregular and poorly sorted. Large pockets of clay occur, the clay flakes showing alteration to sericite in a narrow zone adjoining the quartz, is mainly clear of fine inclusions except for fine trains of limonite particles which occur just inside the margins of the quartz grains which formed part of the original cement before diagenesis and growth of the quartz grains fused them together. However, in places clay still forms the main cement. Quartz gneiss grains occur of similar size range to the quartz mineral grains. To illustrate different provenance of the various quartz mineral grains, some grains are completely clear of strain whereas others show extreme strain. This property is quite independent of grain size. Occasional worn zircon grains are seen in the matrix.

Very little sulphide mineralisation is seen. Occasional complex grains showing irregular bornite and covellite exsolution (?) in chalcopyrite (av. 120μ) - evidence for syngenetic gel precipitation of sulphides in situ or transported with clay in a slurry. Mostly scattered chalcopyrite (av. 10 to 20μ or less) is present. The sulphides occupy the interstices of the quartz grains

and have apparently replaced part of the clay matrix.

A rough modal analysis (v/v percent) is given. Quartz 96, clay mineral 3, zircon <1, sulphide <1.

A comparison of the Lithic Sandstone (2A) with the Red Quartzite (2B) shows :-

- i) sulphide in both specimens shows syngenetic precipitation textures followed by some diagenetic replacement of gangue minerals, mainly interstitial clay; but limited replacement of quartz grain margins only;
- ii) specimen 2B shows a much greater variation in original quartz grain size, the grains in both 2A and 2B tending to be angular to subangular in shape;
- iii) clay mineral content somewhat less for sample 2B; and
- iv) both samples have been subjected to diagenesis about equally.

Specimens 3A, 3B and 4 - taken from the NW corner of the open pit, 50 feet from samples 2A and 2B and at approximately similar stratigraphic horizons in the Red Quartzite and Lithic Sandstone units.

Specimen 3A (20890mN 6774mE) Lithic Sandstone - WYALLA SS

The quartz grains have a greater maximum size (av. 1.5 mm) with a broad size range (down to 20 μ). The larger grains are well rounded and the fine material angular to subangular. Clay cement is more evenly distributed around the quartz grains with fewer fused mosaic patches. Rock fragments of quartz grains and some chert (to .3 mm) occur and also a proportion of feldspar (microcline) grains (to .4 mm) appearing to show some decomposition to sericite. Fine limonite particle trains traverse some of the larger quartz grains indicating former grain boundaries. Occasional worn purple zircons are present in the matrix and traces of rutile.

This specimen is different from the lithic sandstone 2A taken from a closeby location in containing more rounded quartz grains, more chert and rock fragments, some feldspar and in being more poorly sorted. x

Sulphides present are masses of tiny framboidal pyrite grains showing raspberry aggregates of tiny pyrites (separated by carbonaceous envelopes ?) and a porous colloform external zone, up to 60 μ . Sometimes colloform porous masses occur as external 'shells' around quartz grains (rolling pebble action in sulphide mud ?); some pyritohedra forms are seen where framboidal pyrite has recrystallised - post depositional diagenesis. Traces of covellite and chalcocite (10 to 20 μ) occur as complex intergrowths with clay particles. Evidence of syngenetic precipitation of colloidal sulphide with clay or brought in with epigenetic flood material as previously described.

An approximate modal composition (v/v percent) shows quartz 84, chert and rock fragments 5, clay 5, microcline 2, sulphides (mostly pyrite) 4.

WHYALLA SS.
Specimen 3B (30890mN 6774mE) - Lithic Sandstone at junction of red-brown and pale areas. This sandstone is very similar to the previous sandstone examined (specimen 3A) except for a higher proportion of chert and quartz rock fragments (mainly gneiss).

The darker zone is different from the lighter zone only in containing more fine limonite particles in clay matrix adjoining the quartz grains or as a narrow perimeter just inside the quartz grains indicating some diagenetic growth of the quartz; and also by the presence of limonite stained cherts in the dark zone. Large zircon grains (to .2 mm) occur embedded in cherts or in the matrix).

No sulphides are seen in either light or dark zones. The lighter zone may indicate ground water leaching (high limonite in dark zone). This would be borne out by the presence of supergene sulphides at a lower level in the series.

An approximate modal analysis is given here:-

Quartz mineral grains 70, quartz rock fragments 10, chert 15, clay matrix 5, zircon and tourmaline <1.

Specimen 4 (20886mN 6776mE) Red Quartzite ore - PANDURRA FM.

Similar to specimen 2B but quartz grains are rather less angular and fairly equigranular with an average size of 0.3 mm, and there is less interstitial clay. Quartz grains are largely fused into mosaics leaving the usual trails of limonite particle trains inside enlarged quartz grains. The limonite trails indicate that before diagenesis, grains were more rounded than at present. Clay occurs as fine flakes relegated to pockets, altering to sericite. A little detrital tourmaline and zircon occur some of which encroach on and are partially enclosed by growths of diagenetic quartz, the original clay matrix having been expelled by growing quartz. Some crystals of lepidocrocite (av. .4 mm) occur in the clay matrix.

Massive sulphide as chalcocite occurs as veins, 0.6 mm to 3.0 mm wide or as individual grains (av. 0.10 mm size). Within the chalcocite are seen residual bornite and digenite as trains lengthwise along the vein and as complex disordered intergrowths. This would suggest supergene enrichment by chalcocite around original primary bornite (and chalcocite ?). The chalcocite appears weakly anisotropic where blotchy (twinning ?) occurs.

An approximate modal analysis is as follows:-

Quartz mineral grains 82, quartz rock fragments and chert 5, clay 3, sulphides 10, zircon and tourmaline <1 (but more plentiful than in previous sections examined of this series).

Specimens 5 and 5A, adjoining, about 650 feet to the south-east of specimen 4.

Specimen 5 (20812mN 6816mE) Red Quartzite PANDURRA FM.

Similar to specimen 2B but with rather less clay in the interstices and then in pockets; the quartz is more equigranular and therefore there is some greater similarity with specimen 4.

Sorting into bands is evident. Intense diagenesis has occurred which has given quartz grains a jig-saw appearance in mosaics. Trains of limonite particles within quartz grains indicate primary quartz was more rounded. There is a general lack of fine interstitial quartz. Few chert or quartz rock fragments are identifiable. 'Clay' takes on a faint greenish appearance with bluish anomalous low order interference colours in polarised light with crossed nicols.

Traces of chalcopyrite occur but there is more chalcocite (.04 to .1 mm) with clay in the interstices. Approximate 1 mm width veins of chalcocite are seen as cement between some quartz grains. Other chalcocite grains scattered through the section average 300 μ approx. in size.

Chalcocite in bands is most probably supergene and replaced original primary bornite, chalcopyrite or other primary sulphides but this is not evident.

An approximate modal analysis:-

Quartz mineral grains 92, quartz rock fragments 2, clay 3, tourmaline <1, zircon (worn detrital) 1, sulphide 2.

Specimen 5A (20812mN 6816mE). ^{W44/ALLA SS.} Lithic sandstone immediately above Red Quartzite. Large rounded quartz grains and some quartz gneiss rock fragments (av. 1 mm) with a second generation (?) smaller rounded quartz grains in the interstices (av. 0.1 mm). Large rounded cherts occur and large pockets of clay mineral; some clay altering to sericite somewhat limonite stained is plentiful occupying much of the interstices between the large quartz grains. Sericite forms a fine skin around the large quartz grains.

Scattered worn detrital purple zircon and green tourmaline grains occur with the clay. Sulphide as chalcocite faintly anisotropic, (maximum 0.8 mm, av. 0.5 mm) is plentiful and of irregular shape, showing some replacement of clay grains between the quartz grains and margins of some of the grains. Could be the product of diagenesis, it is difficult to say.

Quite a proportion of chalcocite is between 20 and 40 μ approx. and much of this could be lost to slimes in milling.

Modal analysis : quartz mineral grains 72, quartz rock fragments 12, chert 6, clay 5, sulphides 5, tourmaline + zircon <1.

Specimens 6A, B and C (10716mN 6848mE) at the south end of the open pit.

Specimen 6A - Lithic Sandstone. Much like specimen 3A texturally. Like specimen 5A but with fewer silt-size quartz grains in the matrix. Covellite is the main rock cement here varying between 20 and 200 μ thick in the interstices. Some apparent replacement of clay minerals by sulphide is due to post depositional diagenesis. Remaining clay is yellowish limonitic or green, is associated with tiny sub-angular to rounded quartz grains and provides the balance of the rock cement. The larger quartz grains average 1 mm in size, the residual small quartz grains averaging 0.2 mm. Quartz rock fragments and chert are rounded and average about 0.5 mm in size although there are some large quartz gneiss fragments around 1 mm. Some of the chert fragments are quite limonitic. Heavy minerals (tourmaline and zircon) are sparse.

There is a close association here between covellite and interstitial clay and silt rock cement. However, primary covellite is not usual and on the Copperbelt it is most frequently secondary after chalcopyrite. The evidence points to sulphides and matrix being slumped in together with silt and clay prior to consolidation. Rejection of sulphides as well as gangue minerals is shown by clear outer zones of secondary quartz around large quartz grains and local fusing of quartz mosaics. Diagenetic crystallisation tends to obscure original rock/sulphide relationships.

An approximate modal analysis (v/v percentages):-
Quartz mineral grains 63, quartz rock fragments 17, chert 5, clay 5, sulphides 10, tourmaline + zircon <<1.

Specimen 6B (20716mN 6848mE) - Red Quartzite *PANDURRA FM.*

Very similar texturally to specimen 2B. Mainly rounded to subangular quartz grains averaging nearly 1 mm in size. Many of the grains appear somewhat flattened but with no definite orientation and are clear except for narrow trains of fine limonite trains just inside quartz, denoting position of original limits of grain boundaries. A few large quartz rock fragments also occur, some being crowded with fine limonite particles. Very few small quartz grains occur in the clay matrix which is clear and colourless. Occasional olive-green tourmaline grains with colourless authigenic overgrowths occur mainly in the matrix but some are included in margins of post depositional quartz diagenetic growths. Purple rubbly zircons and a little sphene are seen.

Grain sizes are similar to 6A - Lithic Sandstone but in 6B there is more clay matrix and the primary quartz grains are not so rounded.

Sulphides present are seen to be occasional anhedral chalcopryite grains, maximum size 45μ approx., having a broad perimeter of covellite, probably due to post depositional replacement of primary chalcopryite by percolating copper charged solutions before diagenesis sealed-off many of the rock pores.

Approximate modal analysis, v/v percentages:-
Quartz mineral 71, quartz rock 20, chert 3, clay 5, tourmaline + zircon <<1, sulphide <1.

Specimen 6C (20716mN 6848mE) - Red Quartzite *PANDURRA FM.*

A very poorly sorted rock, somewhat similar to 6B but in this section occur areas of finer lenticular quartz (av. 0.3 mm) which have been assimilated into larger quartz grains (av. 0.8 mm). Strained and unstrained quartz grains occur - some maintain optical continuity after diagenetic growth. There are also great variations in quartz grain size. Occasional large quartz grains contain scattered fine rutile needles, are otherwise clear except for the trains of fine limonite inclusions which indicate former grain

boundaries of original constituent quartz grains. Large pockets of clay matrix occur and these are free from quartz inclusions. Quartz rock fragments (mostly quartz gneiss) are sparse; some chert aggregates contain many fine limonite particles. Large zircons appear euhedral and show growth zones. Possible traces of pleochroic andalusite are present.

Sulphide as chalcocite is moderately abundant and occurs as small grains (av. 0.1 mm) occupying some spaces between quartz. One vein of supergene chalcocite (width 0.5 mm) has few inclusions but is intergrown with secondary quartz in veins. Euhedral quartz forms occur lining the secondary veins - grow into chalcocite mass.

Approximate modal analysis, v/v percentages:-

Quartz mineral 87, quartz rock 5, chert 2, clay 5, heavy mineral <1, sulphide 1.

4.1.2 Main Open Pit

Specimen 9A (8402mN 6740mE) - Lithic Sandstone (Protoquartzite of Pettijohn).

Texturally, intermediate between specimen 2A on the one hand and specimens 3A and 6A Lithic Sandstone of West Lagoon with a similar degree of rounding to Specimen 2A.

Comparative petrology possible only where deposition of 'oxide' copper minerals has not obscured the 'sandstone' fabric.

Poorly sorted quartz grains are subangular to rounded and average 0.8 mm in size. Few quartz rock fragments, (quartz gneiss mostly, or chert) occur. The usual trains of fine limonite particles traverse across and form a perimeter just inside the quartz grains demarcating primary quartz grain boundaries. Clay showing some initial recrystallisation to sericite occurs as narrow interstitial zones around and pockets adjoining the quartz grains, the sericite usually forming a thin skin around the quartz. Fine platelets of clay and flakes of sericite occur in channels but these grade into

coarser platelets of chalcedony in pockets.

Some tourmaline prisms with rounded dark green centres, and tiny purple detrital zircons are embedded in authigenic quartz, or in the clay matrix.

The approximate mode of 'non-mineralised' fraction:-
Quartz mineral grains 91, quartz rock fragments 5, chert <1, clay/sericite 3, tourmaline and zircon <1.

The part of the specimen containing oxide copper mineralisation shows unoriented stellate acicular groups of prisms showing high birefringence, faint pleochroism, and oblique extinction which were identified as malachite. The malachite is set in a matrix of chert debris stained with limonite and in areas of a clear pale green 'clay' identified by X-ray diffraction as atacamite. There does appear to be an increase in the ratio of quartz rock fragments to 10 volume percent in the oxide copper zone but this may be a local characteristic.

In one fissure in the rock (caused by the growth of high molecular bulk malachite from low volume sulphides) acicular malachite is seen to line the fissure, the latter being filled in with atacamite as clear green platelets showing anomalous interference colours. In a polished section under low power magnification a 'vein' of pale blue translucent chrysocolla becomes darker when it envelopes the clear dark green transparent atacamite. There exists a very pale green translucent chrysocolla at the junction of the two minerals.

Specimen 9B (8402mN 6740mE) - Red Quartzite *PANDURRA PM.*

Two zones with a sharp contact parallel to the bedding are seen here. One zone consists of rounded grains of quartz (av. 0.3 mm). Cement between the quartz grains consists of limonite stained clay. The other zone, (texturally reminiscent of specimen 2B where not mineralised) contains the copper oxide mineralisation. The quartz grains in the second zone are mainly coarse (av. 0.8 mm to maximum 3 mm) and angular with associated chert and amorphous bluish chrysocolla containing pockets of acicular malachite. The chert,

chrysocolla and malachite generally occupy fissures in the coarse sandstone, the chrysocolla and chert providing the main matrix. Also present in this zone are large zircon grains (av. .1 mm) as euhedra embedded in the matrix and present in appreciable amounts. The appearance of the second zone is not unlike that of a mineralised greywacke or cataclasite.

Some 'geodes' in the mineralised zone are seen to contain botryoidal masses of chrysocolla.

Amorphous looking malachite appears to coat some of the quartz grains of the mineralised zone.

Another section of specimen 9B shows coarse subangular to rounded quartz grains to 2 mm, the spaces between the quartz grains being filled with mainly malachite of colloform appearance and containing 'islands' of chalcedony intergrown with pale blue chrysocolla. In addition both minerals frequently line cavities containing a transparent green flakey mineral, atacamite. Fractures traversing the quartz grains are also filled with malachite as extensions of the main malachite matrix.

Some quartz gneiss fragments occur alternating with the large quartz mineral grains.

In one area of the section, branching veins (0.3 mm in thickness) of chrysocolla, pale blue-green and clear, are margined by in-growing acicular malachite needles. One adjoining cavity is filled with amorphous chrysocolla surrounded by a narrow zone of chalcedony embedded in massive malachite.

Locally, in the malachite are seen pockets of radiate needles of limonite-stained green atacamite showing straight extinction and moderate birefringence. Abundance of copper oxide minerals in Specimen 9B indicate malachite > chrysocolla > atacamite.

Paragenesis of oxide copper minerals appears to depend upon local conditions of carbonated water, colloidal silica in the copper containing waters, or possibly on the presence of dolomite or calcite on the one hand or of chalcedony in the matrix of the sandstone which slowly react with the copper solutions percolating through the mainly clay matrix of the sandstone.

No evidence of the previous presence of sulphides are seen in these rocks such as relict nuclei of chalcocite or residual iron pan horizons from decomposed chalcopyrite or bornite. However, such sulphide relicts have been found and it is assumed that the bulk of oxide copper minerals are not far removed from the original sulphide source and that many of them are the result of in situ replacement due to a radical change in the pH-Eh environment.

Other specimens 10A, B and C were taken from part of the Main Open Cut already samples for metallurgical test work by R. Dudgeon at Mount Gunson Mine.

Specimen 10A (8329mN 6778mE)

A coarse grained sandstone where the quartz grains (averaging 0.2 mm) are mostly subangular; or show rounding due to corrosion or replacement by oxide copper minerals. Cement between the grains is largely malachite which in patches grades into a clear green flakey atacamite mainly isotropic but also displaying anomalous interference colours (amorphous and crystalline material ?) and having RI's significantly higher or about equal to neighbouring quartz grains and possibly lower than those of malachite. The atacamite appears to be replacing the malachite present leaving irregular shaped relict islands of the malachite. Both the atacamite and the malachite traverse fractures in the quartz grains; the atacamite traversing in broad sub-parallel fractures across the quartz mosaics. Large clusters of malachite as radiating fibres display high birefringence. Some malachite with limonite staining which obscures the optical characteristics, occupies fissures in the rock. The malachite contains blue-green islands of isotropic to near isotropic chrysocolla.

Fracture fillings of malachite with associated atacamite and some chrysocolla would not bear milling with an average width of approximately 16 microns and would slime. Larger 'patches' of malachite (av. 2 mm) would not have to be ground excessively to free them from quartz gangue. The matrix component between quartz grains is very high due to previous atmospheric leaching of primary quartz in fractures in an oxidation environment leaving widened channels and pores where copper carbonates and silicates were formed by reaction of permeating copper solutions with colloidal silica and carbonate solutions. These were therefore filled with oxide copper minerals and debris from adjoining leached rocks which also supplied the copper from leached sulphides.

Specimen 10B (8329mN 6778mE)

A poorly sorted sandstone of large, rounded to subangular quartz grains varying between 0.1 and 2.0 mm; some crude banding is shown where maximum grain size decreases to 0.3 mm; grain size is too irregular to obtain a satisfactory average size for the grains. Chert fragments are ten percent of all of the quartz.

Cement and fracture-fill is mainly chrysocolla (blue-green) but also patchily malachite in finer fractures and with pale green atacamite and chalcedony in places. Mineralisation is confined to horizons where there is greatest variation in grain size.

Rounded grains of zircon associated with brown opaque detritus, also perfect tetragonal grains (average 0.1 mm) occur in the interstices.

Milling problems as for Sample 10A but here most malachite is interstitial to quartz and thickness of the cement is 20 μ approx. Milling, sufficiently to free from quartz attachments would cause sliming of the malachite.

Sample 10C (8329mN 6778mE)

Similar to sample 10A, quartz grains average 0.2 mm in size but malachite is more abundant, filling voids up to 0.25 mm and

alternating with clear grass-green glassy atacamite displaying anomalous interference colours and first order reds and yellows. Brownish translucent glass commonly forms nuclei for radiate masses of malachite prisms. Alternating with the patches of green atacamite are colloform masses of chalcedony intergrown with chrysocolla and faintly brown to pale green in colour. Also cross-fibres of chrysocolla form a perimeter around green atacamite in voids. Chrysocolla which lines the vesicles, contains an intergrowth of translucent malachite and clear green atacamite.

As in the previous specimen 10A a zone of fractured chert and quartz occurs containing a matrix of clay or green atacamite. In places atacamite is seen to be intergrown with some malachite. Malachite ensheaths chrysocolla in patches. Perfect hexagonal basal sections of quartz (the result of diagenetic growth) are seen in part of the section containing much limonitic debris.

As in previous sections, quartz grains display cross fractures in which malachite is precipitated.

Encroachment by chrysocolla along fractures following original grain boundaries in constituent quartz-gneiss grains of rock fragments is seen.

One interesting character of this specimen is a transition of variable copper composition from colourless chalcedony, through bluish-white chalcedony to blue-green chrysocolla. This indicates precipitation of silica gel in cavities left by leached quartz and differential absorption by the gel of permeating copper solutions from weathered, oxidised sulphides. The meteoric waters involved were fairly alkaline. Paragenesis of the oxide copper minerals appears generally to be chrysocolla, malachite and then atacamite but continuous variation in the chemical environment will reverse the chrysocolla-malachite sequence.

4.1.3 Mill Feed

A number of thin and polished sections (grains mounted in araldite) taken from Gunson mill feed sampled 16th July 1971 at 8.20 am, were examined.

The thin-section shows quartz grains (approximately .5 to 1 mm) and fragments of quartz mosaics having individual composite grains approximately 1 mm; also quartz gneiss composites (to 1 mm). Crushing has not freed all composite quartz from mosaics where the cement is clay and/or sericite.

Polished sections show a number of interesting composite sulphide grains but it is significant that most sulphides are free from gangue attachments except for covellite which shows complex growths with the gangue mineral. Covellite grain size is frequently $< 5\mu$.

The greatest order of abundance of sulphide concentrations are chalcocite/covellite intergrowths, the former mineral -quite white in reflected light - is pock marked by tiny ribbon-like inclusions of the latter. There are also numerous mottled grains of chalcocite and digenite, intergrown.

Chalcopyrite is the next most abundant sulphide. It is largely free from intergrowth by other sulphides and by gangue. Some grains do show complex intergrowths with pyrite.

The third most abundant sulphide is covellite, complex-locked to gangue (mostly clay), or replacing pyrite as described above.

Pyrite as small euhedral forms (cubes) are seen.

Bornite appears to occur infrequently as a secondary product from the decomposition of chalcopyrite.

There are a number of chalcopyrite or bornite grains

showing attachment to minerals with higher Cu:Fe ratio indicating either supergene enrichment by percolating copper solutions through pores and fissures in the sandstone or simultaneous crystallisation from sulphide gels with subsequent separation of sulphides on crystallisation in ratios dependant upon Cu:Fe:S availability. Both types of reaction occur in the ores at Gunson.

In other polished sections chalcopyrite grains are seen to outnumber chalcocite grains and shows the inhomogeneous nature of ore being mined.

Size of the sulphide copper grains varies 40 to 100 μ for chalcocite and chalcopyrite and the various bornite/digenite/chalcocite/covellite/chalcopyrite intergrowth to approximately 5 μ for covellite. Pyrite is very variable in size from 5 to 150 μ but normally closer to 5 μ . In the larger size range it is usually combined in complex form with chalcopyrite.

In the actual polished sections of samples taken from West Lagoon Open Pit, specimens are seen to be mainly of one kind of sulphide but if the paragenesis can be studied by comparing odd composi sulphide grains it would appear to be gangue $\rightarrow$ pyrite $\rightarrow$ chalcopyrite $\rightarrow$ bornite/covellite $\rightarrow$ chalcocite/digenite/covellite.

4.2 Chemistry

4.2.1 Discussion

Analyses of selected samples from the series were performed by the C.S.R. Central Laboratory. The methods for determining chalcocite copper, 'oxide' copper and chrysocolla copper were originally worked-up by the Central Laboratory for Gunson ores. Additionally the laboratory was asked to determine carbonate to confirm the presence of malachite; phosphate to indicate the possible presence of pseudomalachite; insoluble chloride to determine atacamite; and alumina to determine the presence of kaolinite.

There was insufficient analysed copper to satisfy all determined anion component. As no pseudomalachite was positively identified it was assumed that phosphate present is unidentifiable calcium phosphate associated with clay cement.

When chrysocolla was determined it was almost certainly complex-locked with malachite, and the composite grains floated on the tetrabromethane used as heavy media and which were determined as chrysocolla only. No alkaline earth carbonates were seen in thin section so that carbonate determined was certainly malachite in samples 12, 13A and 13C.

The reported chalcocite copper on samples 1, 2A, 4, 5, 5A, 6A, 6B and 6C unfortunately included chalcopyrite (quite appreciable in samples 2A, 2B and 6B) and other copper, copper/iron sulphides present. No particular sulphide mineral could be said to be specific to Lithic Sandstone or to Red Quartzite owing to post-deposition supergene diagenetic alteration of primary sulphides in both stratigraphic units.

The modal values (volume percent) of minerals given in the Petrography section 4.1 above are very approximate and were not given for specimens 9A, 9B and 10 where copper mineral proportions vary from thin/polished section to section of the same sample.

Comparing the modal values with calculated normative values from chemical composition, they are seen to be grossly at variance even if the adjustment is made to allow for differences in specific gravity of the minerals:-

- i) The high result for kaolinite 8.2 percent in specimen 4, (Table 2), not seen to such a degree in thin section, where it was estimated at 3 percent indicates that feldspar or other aluminium silicates are present. However careful perusal of the section shows kaolinite to be the only aluminium mineral present. Recalculating volume percent to weight percent changes significantly

only quartz:sulphide ratios.

- ii) Differences between modal estimates and normative calculations of sulphides are mainly due to the heterogeneous distribution (as in coarse bands, specimen 4) and also to microscopical observation in only one plane of the polished or thin section. To compare sulphides determined modally with those calculated from normative compositions it is sufficient in these rocks to multiply volume percentages of sulphides by 2.
- iii) The shortcomings of the chemical method for chrysocolla have already been discussed. The atacamite normative composition of specimens 9B, 10A and 10C are far too low compared with amounts seen in thin section.

4.2.2 Conclusions

The chemical analyses only confirm the presence of oxide copper minerals known to be present from microscopy and X-ray diffraction. Some analyses (e.g. atacamite where determined) are so grossly at variance with amounts seen in thin and polished section as to indicate that further development of analytical methods might be necessary.

The above comments apply only to those empirical methods recently developed by the Central Laboratory for determining copper due to chalcocite, chrysocolla, etc., and does not apply to routine assay methods.

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TABLE 1

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MT. GUNSON COPPER ORES, CHEMICAL ANALYSES (W/W%) BY CSR CENTRAL LABORATORY

Sample No.	Field No.	Sample	Total Cu	Total Fe	'Oxide' Cu	Chrysocolla Cu	Malachite Cu	Chalcocite Cu	Al ₂ O ₃	PO ₄ ³⁻	Insol. Cl
A 4000		Mt. Gunson Mill Feed 8.20 am, 15/7/1971	0.66	1.03	0.10	-	-	0.56	-	-	-
A 4001	2A	Lithic Sandstone West Lagoon	0.35	0.76	0.01	-	-	0.34	-	-	-
A 4005	4	Red Quartzite Ore West Lagoon	4.38	0.09	0.33	-	-	4.05	3.23	-	-
A 4006	5	Red Quartzite Ore West Lagoon	2.69	0.18	0.23	-	-	2.46	-	-	-
A 4007	5A	Lithic Sandstone immed., above Red Quartzite, West Lagoon	9.13	0.39	0.47	-	-	8.66	2.23	-	-
A 4008	6A	Lithic Sandstone West Lagoon	6.66	0.11	0.83	-	-	5.83	2.04	-	-
A 4009	6B	Red Quartzite West Lagoon	0.32	0.06	0.05	-	-	0.28	-	-	-
A 4010	6C	Red Quartzite West Lagoon	1.01	0.06	0.21	-	-	0.80	-	-	-
A 4012	9B	Red Quartzite Main Open Cut	8.34	-	-	6.41	0.87	N.D.	-	0.10	0.3
A 4013A	10	Metallurgical Test Area	10.31	-	-	8.51	0.77	N.D.	0.54	0.15	0.3
A 4013C	10	"	14.94	-	-	14.27	0.57	N.D.	-	0.05	0.4

N.D. : Not detected

'Oxide' Cu : Acid soluble copper (non-sulphide) and will include about half Cu O present.

Chrysocolla Cu : TBE float but included originally, malachite locked to crushed chrysocolla grains.

Malachite Cu : Determined from carbonate analysis as no alkaline earth carbonates were seen in thin-section.

Chalcocite Cu : Could include other sulphides present such as chalcopyrite, digenite, bornite and covellite and also half of any tenorite present.

TABLE 2

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MT. GUNSON COPPER ORES, CALCULATED MINERAL COMPOSITIONS (FROM CSR CENTRAL LABORATORY ANALYSES)

Sample No.	Field No.	Sample	Chalcocite Cu_2S	Malachite $\text{Cu}_2(\text{OH})_2\text{CO}_3$	Chrysocolla $\text{CuSiO}_3 \cdot 2\text{H}_2\text{O}$	Atacamite $\text{CuCl}_2 \cdot 3\text{Cu}(\text{OH})_2$	Kaolinite $\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$	Calcium Phosphate $\text{Ca}_3(\text{PO}_4)_2$
A 4000	1	Mt. Gunson Mill Feed 8.20 am 15/7/1971	0.7	-	-	-	-	-
A 4001	2A	Lithic Sandstone West Lagoon	0.4	-	-	-	-	-
A 4005	4	Red Quartzite Ore West Lagoon	5.1	-	-	-	8.2	-
A 4006	5	Red Quartzite Ore West Lagoon	3.1	-	-	-	-	-
A 4007	5A	Lithic Sandstone above Red Quartzite, West Lagoon	10.8	-	-	-	5.6	-
A 4008	6A	Lithic Sandstone West Lagoon	7.3	-	-	-	5.2	-
A 4009	6B	Red Quartzite West Lagoon	0.4	-	-	-	-	-
A 4010	6C	Red Quartzite West Lagoon	1.0	-	-	-	-	-
A 4012	9B	Red Quartzite Main Open Cut	-	1.5	17.7	1.8	-	0.16
A 4013A	10A	Metallurgical Test Area	-	1.3	23.5	1.8	1.4	0.24
A 4013C	10C	Main Open Cut	-	1.0	39.4	2.4	-	0.08